Algebraic numbertheory
exSheet 10

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Points 21.5/22
(97.73%)
Well Done!

Aufgabe 1

Let $p \in \mathbb{Z}$ be a prime number.

t.s. There exist $a, b \in \mathbb{Z}$ such that

$$a^2 + b^2 \equiv -1 \pmod{p}.$$

proof. From lecture we know that

$$\#\{x^2 : x \in \mathbb{F}_p^\times\} = \frac{p-1}{2}.$$

Then we define $A = \{x^2 : x \in \mathbb{F}_p\}$, $B = \{-1 - y^2 : y \in \mathbb{F}_p\}$.

Since p is prime number, we get that $\mathbb{F}_p$ is a integral domain thus $0 \notin \{x^2 : x \in \mathbb{F}_p^\times\}$ ($x^2 = 0 \Rightarrow x = 0$ & $0 \notin \mathbb{F}_p^\times$) but $0 \in A$. ($0^2 = 0$) Hence, with $A = \{x^2 : x \in \mathbb{F}_p^\times\} \cup \{0\}$ we get

$$\#A = \#\{x^2 : x \in \mathbb{F}_p^\times\} + 1 = \frac{p-1}{2} + 1 = \frac{p+1}{2}. \checkmark$$

claim $\#A = \#B$.

This claim implies that $\#B = \frac{p+1}{2}$. But since $A, B \subseteq \mathbb{F}_p$ by construction and

$$\#A + \#B = \frac{p+1}{2} + \frac{p+1}{2} = p + 1 > p = \#(\mathbb{F}_p),$$

A, B can not be disjoint. Hence, there exist $\psi \in A \cap B$ $\Big\}$ Thus there exist $x, y \in \mathbb{F}_p$ such that

$$x^2 = \psi = -1 - y^2 \quad \text{hence} \quad x^2 + y^2 = -1.$$

Then by taking any $a, b \in \mathbb{Z}$ such that $\pi(a) = x$, $\pi(b) = y$, where $\pi : \mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ canon. projection (which is surjective), it holds that

$$-1 = x^2 + y^2 = \pi^2(a) + \pi^2(b) = \pi(a^2) + \pi(b^2)$$

thus $a^2 + b^2 \equiv -1 \pmod{p}$ and we can conclude.

It is left to prove the claim.

We define $\psi : \mathbb{F}_p \to \mathbb{F}_p$, $x \mapsto -1 - x$. Then $\psi_1 : \mathbb{F}_p \to \mathbb{F}_p$, $x \mapsto -x$ is a Grouphom ($\mathbb{F}_p$ as additive group, $-(a+b) = -a - b$) and since $\ker(\psi_1) = \{0\}$ and $\mathbb{F}_p$ finite it is a group iso in part. bijective. Then $\psi_2 : \mathbb{F}_p \to \mathbb{F}_p$, $x \mapsto 1 + x$ is a bijection as well since $\mathbb{F}_p \supset \mathbb{F}_p$ ($x \mapsto x - 1$ its inverse). Since $\psi = \psi_1 \circ \psi_2$ it follows that ψ is a bijection. Now by construction of ψ, we get that

$$B = \psi(A)$$

Since ψ a bijection claim follows and we can conclude.

□ 3

Aufgabe 2

Let $b \in \mathbb{Z}$ be positive with $b = p_1 \cdot \ldots \cdot p_g$ its prime number decomposition.
For $a \in \mathbb{Z}$, we define

$$\left(\frac{a}{b}\right) = \prod_{i=1}^{g} \left(\frac{a}{p_i}\right)$$

(i) We assume b being odd.

t.S. $\left(\frac{a+kb}{b}\right) = \left(\frac{a}{b}\right) \quad \forall a, k \in \mathbb{Z}$.

proof. Let $a \in \mathbb{Z}$ and let $k \in \mathbb{Z}$.
By definition, it holds that

$$\left(\frac{a+kb}{b}\right) = \prod_{i=1}^{g} \left(\frac{a+kb}{p_i}\right) \quad \checkmark$$

And by assumption, b is odd thus $p_i > 2$ for each $i \in \{1, \ldots, g\}$, thus p_i odd.
Furthermore, it holds for any $i \in \{1, \ldots, g\}$ that $p_i \mid b$ thus $p_i \mid kb$
hence, $a + kb \equiv a \pmod{p_i}$ and Lemma 3.30 provides that

$$\left(\frac{a+kb}{p_i}\right) = \left(\frac{a}{p_i}\right) \quad \forall i \in \{1, \ldots, g\}.$$

This implies then

$$\left(\frac{a+kb}{b}\right) = \prod_{i=1}^{g} \underbrace{\left(\frac{a+kb}{p_i}\right)}_{=\left(\frac{a}{p_i}\right)} = \prod_{i=1}^{g} \left(\frac{a}{p_i}\right) \overset{\text{Def}}{=} \left(\frac{a}{b}\right)$$

and we can conclude. ☐ (2)

Aufgabe 2

(ii) t.s. $\forall a,b \in \mathbb{Z}: a \geq 1, b \geq 1$ odd, it holds that

$$\left(\frac{a}{b}\right) = (-1)^{\frac{a-1}{2}\cdot\frac{b-1}{2}} \left(\frac{b}{a}\right)$$

In order to prove this statement, we prove a lemma before

Lemma Let $p_1, \dots, p_n \in \mathbb{Z}$ be some odd, positive integers, for $n \in \mathbb{N}$.

Then

$$\sum_{i=1}^{n} \frac{p_i - 1}{2} \equiv \frac{p_1 \cdot \dots \cdot p_n - 1}{2} \pmod 2$$

proof. First, let $\pi: \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z}$ be the canonical projection.

Thm by assumption, for any $i \in [1, \dots, n]$ there exists a $a_i \in \mathbb{Z}$ s.t.

$p_i = 2a_i + 1$.

Then it holds for $n=1$ that $\frac{p_1 - 1}{2} \equiv \frac{p_1 - 1}{2} \pmod 2$ (thus the statement holds for $n=1$)

Assuming the statement holds for a $n \in \mathbb{N}$.

Then we have

$$\pi\left(\frac{p_1 \cdot \dots \cdot p_{n+1} - 1}{2}\right) \overset{\checkmark}{=} \pi\left(\frac{p_1 \cdot \dots \cdot p_n (2a_{n+1}+1) - 1}{2}\right)$$

$$\overset{\checkmark}{=} \pi\left(\frac{2a_{n+1} p_1 \cdot \dots \cdot p_n}{2}\right) + \underbrace{\pi\left(\frac{p_1 \cdot \dots \cdot p_n - 1}{2}\right)}_{= \pi\left(\sum_{i=1}^{n} \frac{p_i - 1}{2}\right) \text{ by ind. hypothesis.}}$$

$$\overset{\checkmark}{=} \pi(a_{n+1}) \cdot \underbrace{\pi(p_1 \cdot \dots \cdot p_n)}_{= \pi(p_1) \cdot \dots \cdot \pi(p_n) = 1} + \pi\left(\sum_{i=1}^{n} \frac{p_i - 1}{2}\right)$$

since p_i odd

$$\pi(a_{n+1}) + \pi\left(\sum_{i=1}^{n} \frac{p_i - 1}{2}\right). \qquad \text{(canon. proj. a groupthom.)}$$

But then $\frac{p_{n+1} - 1}{2} = \frac{2a_{n+1} + 1 - 1}{2} = a_{n+1}$ thus it implies that

$$\pi\left(\frac{p_1 \cdot \dots \cdot p_{n+1} - 1}{2}\right) = \pi\left(\frac{p_{n+1} - 1}{2}\right) + \pi\left(\sum_{i=1}^{n} \frac{p_i - 1}{2}\right)$$

$$= \pi\left(\sum_{i=1}^{n+1} \frac{p_i - 1}{2}\right)$$

and we can conclude. ∎

(4)

Aufgabe 2(i)

Now we prove the statement of exercise.

proof. Let $a = q_1 \cdot \ldots \cdot q_h$, $b = p_1 \cdot \ldots \cdot p_g$ be the prime number decomposition. $h, g \geq 1$.

Then it holds that $q_k > 2$, $p_l > 2$ for any $(k,l) \in \underbrace{\{1,\ldots,h\}}_{=:H} \times \underbrace{\{1,\ldots,g\}}_{=:G}$

since a, b are each odd, thus q_k, p_l are each odd.

Then we have by Definition and Lemma 3.30(iii), that

$$\left(\frac{a}{b}\right) \overset{\text{def}}{=} \prod_{i=1}^{g} \left(\frac{a}{p_i}\right) \overset{3.30(iii)}{=} \prod_{i=1}^{g} \prod_{j=1}^{h} \left(\frac{q_j}{p_i}\right).$$

Then we may assume that there exists no $(k,l) \in H \times G$ such that $q_k = p_l$.

Otherwise, we have $p_l = q_k \mid a \overset{\text{def}}{\Rightarrow} \left(\frac{a}{q_k}\right) = 0 \overset{\text{def}}{\Rightarrow} \left(\frac{a}{b}\right) = 0$

and $q_k = p_l \mid b \overset{\text{def}}{\Rightarrow} \left(\frac{b}{p_l}\right) = 0 \overset{\text{def}}{\Rightarrow} \left(\frac{b}{a}\right) = \prod_{\substack{j=1 \\ j \neq l}}^{h} \left(\frac{b}{p_j}\right) \cdot \underbrace{\left(\frac{b}{p_l}\right)}_{=0} = 0$. thus $\left(\frac{a}{b}\right) = \left(\frac{b}{a}\right)$ and

we would be finished. ✓

Hence, we may apply Satz 3.28 which provides us $\forall (j,i) \in H \times G$ that

$\left(\frac{q_j}{p_i}\right) = (-1)^{\frac{p_i-1}{2} \cdot \frac{q_j-1}{2}} \left(\frac{p_i}{q_j}\right)$. So we get

$$\left(\frac{a}{b}\right) = \prod_{i=1}^{g} \prod_{j=1}^{h} (-1)^{\frac{p_i-1}{2} \cdot \frac{q_j-1}{2}} \left(\frac{p_i}{q_j}\right) \overset{\text{3 lemma}}{=} \prod_{j=1}^{h} \prod_{i=1}^{g} (-1)^{\frac{p_i-1}{2}\frac{q_j-1}{2}} \left(\frac{p_i}{q_j}\right)$$

$$= \prod_{j=1}^{h} (-1)^{\frac{q_j-1}{2} \sum_{i=1}^{g} \frac{p_i-1}{2}} \underbrace{\prod_{i=1}^{g} \left(\frac{p_i}{q_j}\right)}_{= \left(\frac{b}{q_j}\right) \; 3.30(iii)}$$

$$= (-1)^{S_1 \cdot S_2} \cdot \prod_{j=1}^{h} \left(\frac{b}{q_j}\right) = (-1)^{S_1 \cdot S_2} \left(\frac{b}{a}\right), \text{ where } \begin{cases} S_1 = \sum_{j=1}^{h} \frac{q_j-1}{2} \\ S_2 = \sum_{i=1}^{g} \frac{p_i-1}{2} \end{cases}$$

Then previous Lemma provides us

$$S_1 \equiv \frac{q_1 \cdot \ldots \cdot q_h - 1}{2} \pmod 2 \equiv \frac{a-1}{2} \pmod 2 \quad ✓$$

$$S_2 \equiv \frac{p_1 \cdot \ldots \cdot p_g - 1}{2} \pmod 2 \equiv \frac{b-1}{2} \pmod 2 \quad ✓$$

hence, $S_1 \cdot S_2 \equiv \frac{a-1}{2} \cdot \frac{b-1}{2} \pmod 2$ and we get

$$(-1)^{S_1 \cdot S_2} = (-1)^{\frac{a-1}{2} \cdot \frac{b-1}{2}}$$

which gives us

$$\left(\frac{a}{b}\right) = (-1)^{\frac{a-1}{2}\frac{b-1}{2}} \left(\frac{b}{a}\right) \quad \text{and we can conclude.} \quad ✓$$

□

4

Aufgabe 3

No this statement does not hold. We consider $a=3$, $b=35$.

claim $\left(\frac{a}{b}\right)=1$ but there exist no $x\in\mathbb{Z}$ s.t. $x^2\equiv a \pmod b$.

proof By definition, it holds that

$$\left(\frac{a}{b}\right)=\left(\frac{3}{5}\right)\cdot\left(\frac{3}{7}\right) \qquad (35=5\cdot 7,\ 5,7 \text{ prime } \in\mathbb{N})$$

Then $1^2\equiv 1,\ 2^2\equiv 4 \pmod 5$ thus $\{\pi_5(1),\pi_5(4)\}\subseteq\{x^2 : x\in\mathbb{F}_5^{\times}\}$ but since $\#\{x^2 : x\in\mathbb{F}_5^{\times}\}=\frac{5-1}{2}=2$, the equality follows.

(where $\pi_5:\mathbb{Z}\to\mathbb{Z}/5\mathbb{Z}$, $\pi_7:\mathbb{Z}\to\mathbb{Z}/7\mathbb{Z}$ canon. proj.)

Then $1^2\equiv 1,\ 2^2\equiv 4,\ 3^2\equiv 2 \pmod 7$ since $\frac{7-1}{2}=3$ we get by the same argument that $\{x^2 : x\in\mathbb{F}_7^{\times}\}=\{\pi_7(1),\pi_7(2),\pi_7(4)\}$

Then since $a=3\in\mathbb{F}_5^{\times},\mathbb{F}_7^{\times}$ ($\mathbb{F}_5^{\times}=\mathbb{F}_5\setminus\{0\}$, $\mathbb{F}_7^{\times}=\mathbb{F}_7\setminus\{0\}$, since $5,7$ prime $\in\mathbb{N}$) we get by definition that

$$\left(\frac{3}{5}\right)=-1,\quad \left(\frac{3}{7}\right)=-1$$

Thus $\left(\frac{a}{b}\right)=1$. ✓

Then we assume that there exists a $x\in\mathbb{Z}$ s.t. $x^2\equiv a \pmod b$ $(a=3,b=35)$ ✓

Then $5\mid 35$ ✓, & $35\mid x^2-3$ ✓ thus $5\mid x^2-3$ since $x^2\equiv 3 \pmod 5$ ✓

But $\pi_5(3)\notin\{x^2 : x\in\mathbb{F}_5^{\times}\}$ ($5\nmid 3$ and $3\not\equiv 1,4 \pmod 5$). Hence there can not exist a $x\in\mathbb{Z}$ s.t. $\pi_5(x^2)=(\pi_5(x))^2=\pi_5(3)$ (π_5 surjective) since, this is by assumption a contradiction and the statement follows. ✓

(2)

Aufgabe 4

Let $p \in \mathbb{Z}$ be a prime numbers with $p \equiv 5 \pmod 8$.

(this assumption is approved by prof. Habegger because of what I mailed to you. (Asked him during the lecture))

(i) Let $n \in \mathbb{Z}$ and let $\ell \in \mathbb{Z}$ prim s.t. $\ell \mid (n^2 + n - \frac{p-1}{4})$, where $p \neq \ell$.

t.s. $\left(\frac{\ell}{p}\right) = 1$.

proof. Since $p \equiv 5 \pmod 8$, there exists a $r \in \mathbb{Z}$ s.t. $p = 5 + 8r$.

claim 1 $\ell \neq 2$.

We prove this claim 1 later.

claim 2. $\left(\frac{\ell}{p}\right) = \left(\frac{p}{\ell}\right)$.

Since $\ell \neq 2$, ℓ is odd, and by precondition $\ell \neq p$. so by Satz 3.28 it holds that

$$\left(\frac{\ell}{p}\right) = (-1)^{s_1 \cdot s_2} \left(\frac{p}{\ell}\right), \text{ where } s_1 = \frac{p-1}{2}, \; s_2 = \frac{\ell - 1}{2}$$

Since ℓ odd, $\frac{\ell-1}{2} \in \mathbb{Z}$, then $\frac{p-1}{2} = \frac{5+8r-1}{2} = 2 + 4r \in 2\mathbb{Z} \Rightarrow s_1 \cdot s_2 \in 2\mathbb{Z}$

Hence, $(-1)^{s_1 \cdot s_2} = 1$ and claim 2 follows. ✓

By assumption it holds that $n^2 + n - \frac{p-1}{4} \in \ell\mathbb{Z}$ thus

$$4n^2 + 4n - p + 1 \in \ell\mathbb{Z} \Rightarrow (2n+1)^2 = 4n^2 + 4n + 1 \in p + \ell\mathbb{Z}.$$

Hence, $(2n+1)^2 \equiv p \pmod \ell$ ✓

By claim 1 ℓ is odd, so we get by Lemma 3.30 (i) & (iii) that

$$\left(\frac{p}{\ell}\right) = \left(\frac{(2n+1)^2}{\ell}\right) = \left(\frac{2n+1}{\ell}\right)^2$$

We claim that $\ell \nmid 2n+1$. Indeed, if $\ell \mid 2n+1$, there exists a $r \in \mathbb{Z}$, s.t. $r \cdot \ell = 2n+1$, thus

$$r^2\ell^2 = (2n+1)^2 \in p + \ell\mathbb{Z}$$

Hence, there exists a $s \in \mathbb{Z}$ s.t. $r^2\ell^2 + s\ell = \ell(r^2\ell + s) = p$.

But since $\ell \neq p$, $r^2\ell + s \neq 1$ but since p is irred. (in $\mathbb{Z}$) This is absurd. ↯

Thus $\ell \nmid 2n+1$ therefore $\frac{2n+1}{\ell} = \pm 1$ which implies that

$$\left(\frac{\ell}{p}\right) \overset{\text{claim 2}}{=} \left(\frac{p}{\ell}\right) = \left(\frac{2n+1}{\ell}\right)^2 = 1$$

It is left to prove claim 1.

We assume $\ell = 2$. Since $\ell \mid n^2 + n - \frac{p-1}{4}$ $\left(= \frac{5+8r-1}{4}\right)$, this implies

$$n^2 + n - \frac{4+8r}{4} = n(n+1) - 1 - 2r \in 2\mathbb{Z}$$

thus $n(n+1) - 1 \in 2\mathbb{Z}$. But since $n(n+1)$ even for any $n \in \mathbb{Z}$ (also work if $n = 0$) ✓

(n odd $\Rightarrow n+1$ even) this is absurd. Hence ℓ can not be 2. ↯

(2)

Aufgabe 4

(ii)

t.s. $\forall n \in \mathbb{Z}$, there exists a $x \in \mathbb{Z}$ such that

$$n^2 + n - \frac{p-1}{4} \equiv x^2 \pmod{p}$$

proof. Let $n \in \mathbb{Z}$.

Then it holds that

$$4\left(n^2+n-\tfrac{p-1}{4}\right) = 4n^2 + 4n - p + 1$$
$$= (2n+1)^2 - p.$$

Thus $4\left(n^2+n-\tfrac{p-1}{4}\right) \equiv (2n+1)^2 \pmod{p}$

Since $2n \neq 1$ thus $2n+1 \neq p\mathbb{Z}$, we get by definition that $\left(\frac{4(n^2+n-\frac{p-1}{4})}{p}\right) = 1$. Since p a prime with $p = \underbrace{s}_{\text{odd}} + \underbrace{r8}_{\text{even}}$ for a $r \in \mathbb{Z}$ it must be odd. So by Lemma 3.30 (iii) we get

$$\left(\frac{4}{p}\right) \cdot \left(\frac{n^2+n-\frac{p-1}{4}}{p}\right) = 1.$$

Then $\left(\frac{4}{p}\right) = \left(\frac{2}{p}\right)^2$. But since p odd $\Rightarrow p > 2$ hence p can not divide 2.

Therfore $\left(\frac{2}{p}\right) = \pm 1 \Rightarrow \left(\frac{4}{p}\right) = 1$, which implies

$$\left(\frac{n^2+n-\frac{p-1}{4}}{p}\right) = 1.$$

By definition of legendre symbol, there exists a $x \in \mathbb{Z}$ such that

$$n^2+n-\frac{p-1}{4} \equiv x^2 \pmod{p}.$$

□

(2)

Aufgabe 5

Let $p \in \mathbb{Z}$ be an odd prime number and $b \in \mathbb{Z}$. $\forall a \in \mathbb{Z}$, we write $\bar{a} = a + p\mathbb{Z} \in \mathbb{F}_p$.

We define

$$E = \{(x,y) \in \mathbb{F}_p^2 : y^2 = x^3 + \bar{b}\}.$$

Furthermore we define $\pi_1, \pi_2 : E \longrightarrow \mathbb{F}_p$ as projections to 1,2 coordinates

$$\pi_1 : (x,y) \longmapsto x, \quad \pi_2 : (x,y) \longmapsto y.$$

(i)

t.s. $\#E = p + \sum_{n=0}^{p-1} \left(\frac{n^3+b}{p}\right)$.

proof. It holds that $\mathbb{F}_p = \bigcup_{n=0}^{p-1} \{\bar{n}\}$

Thus $E = \pi_1^{-1}(\mathbb{F}_p) = \pi_1^{-1}\left(\bigcup_{n=0}^{p-1}\{\bar{n}\}\right) = \bigcup_{n=0}^{p-1} \pi_1^{-1}(\{\bar{n}\})$ which is a disjoint union since $\bigcup_{n=0}^{p-1}\{\bar{n}\}$ is.

Then we define $\forall n \in \{0,\dots,p-1\}: E_n = \pi_1^{-1}(\{\bar{n}\})$. $\Rightarrow E = \bigcup_{n=0}^{p-1} E_n$ disjoint union.

claim $\forall n \in \{0,\dots,p-1\}: \#E_n \le 2$.

Thus $\#E_n \in \{0,1,2\}$. $\forall n$.

Let $n \in \{0,\dots,p-1\}$.

- If $\#E_n = 2$, then there exist distinct $y_1, y_2 \in \mathbb{F}_p$ s.t. $y_1^2 = y_2^2 = \bar{n}^3 + \bar{b} = \overline{n^3+b}$
  But then $(-y_1)^2 = y_1^2 = \overline{n^3+b}$. But since $\#E_n = 2 \Rightarrow -y_1 = y_2$ and with $y_1 \neq y_2 = -y_1 \Rightarrow y_1 \neq 0$. Since p is prime thus $\mathbb{F}_p$ an integral domain it follows that $y_1^2 = \overline{n^3+b} \neq 0$. So we get by definiton of Legendre symbol, that $\left(\frac{n^3+b}{p}\right) = 1$ and therefore $\#E_n = 2 = 1 + \left(\frac{n^3+b}{p}\right)$. ok
- If $\#E_n = 1$, there exists $y \in \mathbb{F}_p$ s.t. $y^2 = \overline{n^3+b}$ but since $(-y)^2 = y^2$ and $\#E_n = 1$ we get $y = 0$ thus $\left(\frac{n^3+b}{p}\right) = 0$ hence, $(y = 0 = \overline{n^3+b} \Rightarrow n^3+b \in p\mathbb{Z})$ ok
  $$\#E_n = 1 = 1 + \left(\frac{n^3+b}{p}\right)$$
- If $\#E_n = 0$, there exists no $y \in \mathbb{F}_p$ such that $y^2 = \overline{n^3+b}$ thus by definition $\left(\frac{n^3+b}{p}\right) = -1$ ok
  which implies
  $$\#E_n = 0 = 1 + \left(\frac{n^3+b}{n}\right).$$

So it holds for any $n \in \{0,\dots,p-1\}$: $\#E_n = 1 + \left(\frac{n^3+b}{p}\right)$. ok Since $\bigcup_{n=0}^{p-1} E_n$ is a disjoint union, we get

$$\#E \overset{\checkmark}{=} \sum_{n=0}^{p-1} \#E_n \overset{\checkmark}{=} \sum_{n=0}^{p-1} 1 + \left(\frac{n^3+b}{p}\right) = p + \sum_{n=0}^{p-1}\left(\frac{n^3+b}{p}\right).$$

It is left to prove the claim.

We define $A = \{\bar{1},\dots,\overline{\frac{p-1}{2}}\}$, $B = \{\overline{\frac{p+1}{2}},\dots,\overline{p-1}\}$. Then $A = -B$, hence and we get $\frac{p-1}{2} = \#\{x^2 : x \in \mathbb{F}_p^\times\} = \#\{x^2 : x \in A\} = \#\{x^2 : x \in B\}$ $(\forall x \in A : \exists y \in B : x^2 = y^2)$.

Then we assume that, these exist $n \in \{0,\dots,p-1\}$ s.t. $\#E > 2$, thus there exist distinct $x,y,z \in \mathbb{F}_p$ s.t. $x^2 = \overline{n^3+b} = y^2 = z^2$. Since x,y,z distinct $x^2 = y^2 = z^2 \neq 0 \Rightarrow x,y,z \neq 0$ ($\mathbb{F}_p$ integrdom.)

if $-x = y \Rightarrow z \neq x$, $z \neq y = -x$, we may assume that $y = -x$ (else repl. y with z).

Furthermore we may assume that $x,y \in A$ (otherwise replace it with $-x,-y$) ok

But then it holds that $\#\{x^2 : x \in A\} < \frac{p-1}{2}$ since $\#A = \frac{p-1}{2}$ since $x,y \in A : x \neq y \Leftrightarrow x^2 = y^2$.

But since $\frac{p-1}{2} = \#\{x^2 : x \in \mathbb{F}_p^\times\} = \#\{x^2 : x \in A\}$, this is absurd. Hence, there can not exist such x,y,z which is by assumption a contradiction. It follows that

$$\#E_n \le 2 \quad \forall n \in \{0,\dots,p-1\}.$$ ok

Aufgabe 5

(ii)

t.s. if $b \in p\mathbb{Z}$, then $\#E = p$.

proof We assume $b \in p\mathbb{Z}$. Then by lemma 3.30 on (i), we get

$$\#E \overset{\text{th}}{=} p + \sum_{n=0}^{p-1} \underbrace{\left(\frac{n^3+b}{p}\right)}_{=\left(\frac{n^3}{p}\right)} \overset{\checkmark}{=} p + \sum_{n=0}^{p-1} \left(\frac{n}{p}\right)^3$$

$$\overset{\checkmark}{=} \left(\frac{n^3}{p}\right) \overset{\checkmark}{=} \left(\frac{n}{p}\right)^3 \text{ Lemma 3.30}$$

We define $A = \{x^2 : x \in \{0, \dots, p-1\}\}$, $B = \{0, \dots, p-1\} \setminus (A \cup \{0\})$.

Then is $\{0, \dots, p-1\} = A \cup B \cup \{0\}$ a disjoint union and we get

$$\#E = p + \sum_{n \in A} \left(\frac{n}{p}\right)^3 + \sum_{n \in B} \left(\frac{n}{p}\right)^3 + \underbrace{\left(\frac{0}{p}\right)^3}_{=0 \text{ by def}}$$

$$\Rightarrow \left(\frac{n}{p}\right)^3 = 1$$

Then by construction of A, B we get $\forall n \in A: \left(\frac{n}{p}\right) = 1$, $\forall n \in B: \left(\frac{n}{p}\right) = -1$, $\left(\frac{n}{p}\right)^3 = -1$.

Furthermore, by identifying $\{0, \dots, p-1\} = \mathbb{F}_p$, we get by $\{x^2 : x \in \mathbb{F}_p^\times\} = \frac{p-1}{2}$

$\Rightarrow$ that $\#A = \frac{p-1}{2}$, $\#B = p - 1 - \frac{p-1}{2} = \frac{p-1}{2}$ hence,

$$\sum_{n \in A} \underbrace{\left(\frac{n}{p}\right)^3}_{=1} = \frac{p-1}{2}, \quad \sum_{n \in B} \underbrace{\left(\frac{n}{p}\right)^3}_{=-1} = -\frac{p-1}{2}$$

So it follows

$$\#E \overset{\text{th}}{=} p + \sum_{n \in A} \left(\frac{n}{p}\right)^3 + \sum_{n \in B} \left(\frac{n}{p}\right)^3 \overset{\checkmark}{=} p + \frac{p-1}{2} - \frac{p-1}{2} \overset{\checkmark}{=} p$$

and we can conclude. ✓

(2)

(iii)

t.s. if $p \equiv 2 \pmod 3$, then $\#E_p$.

proof We assume $p \equiv 2 \pmod 3$.

claim $\mathbb{F}_p \longrightarrow \mathbb{F}_p$, $x \longmapsto x^3$ is a bijection

Then a translation on a finite field by an elm of the field is a bijection $\overset{\text{th}}{=} \left((x \mapsto b+x+b)^{-1} = (x \mapsto x-b)\right)$

Thus $f: \mathbb{F}_p \longrightarrow \mathbb{F}_p$, $x \longmapsto x^3 + b$ is a bijection.

So we can write E as

$$E = \{(x, y) \in \mathbb{F}_p^2 : y^2 = f(x)\}.$$

and $E = \bigcup_{m=0}^{p-1} \pi_2^{-1}(\{\bar{m}\})$ with the same argument as in (i) (this time proj. on 2nd coord)

Then every elm of $\pi_2^{-1}(\{\bar{m}\})$ has the form $(x, \bar{m})$ with $f(x) = \bar{m}^2$.

But f is a bijection thus $\forall m \in \{0, \dots, p-1\}$: there exists an unique $x \in \mathbb{F}_p$

s.t. $\bar{m}^2 = f(x) = x^2 + b$ hence $\{(x, \bar{m})\} = \pi_2^{-1}(\{\bar{m}\})$. and we get

$$\#E = \#\left(\bigcup_{m=0}^{p-1} \pi_2^{-1}(\{\bar{m}\})\right) \overset{\text{disj. un.}}{=} \sum_{m=0}^{p-1} \#(\pi_2^{-1}(\{\bar{m}\})) = \sum_{m=0}^{p-1} 1 = p \checkmark$$

It is left to prove the claim.

Since $\mathbb{F}_p$ is finite, we only need to show the surjectivity of the map.

Let $y \in \mathbb{F}_p$. Since $0 \longmapsto 0^3 = 0$, we may assume that $y \in \mathbb{F}_p^\times$.

From Algebra I, we know that any multiplicative subgroup of a finite field is cyclic. Hence $\mathbb{F}_p^\times$ is cyclic. Therefore, there exist a $g \in \mathbb{F}_p^\times$ s.t. $\langle g \rangle = \mathbb{F}_p^\times$.
Then there exists a $r \in \mathbb{Z}$ s.t. $y = g^r$.

We may assume that 3 divides r.
Because if not, then $r \equiv 1 \pmod 3$ or $r \equiv 2 \pmod 3$

- If $r \equiv 2 \pmod 3$, then $r+(p-1) \equiv 2+1 \pmod 3 \equiv 0 \pmod 3$ since $p \equiv 2 \pmod 3$, hence $3 \mid r+(p-1)$. But then we get by Fermats little thm that

$$g^{r+p-1} = g^r \cdot \underbrace{g^{p-1}}_{=1 \text{ in } \mathbb{F}_p \text{ since } g \neq 0} = g^r = y.$$

and we could replace r with $r+p-1$. and?

- If $r \equiv 1 \pmod 2$? then $r + 2(p-1) \equiv 0 \pmod 3 \Rightarrow 3 \mid r+2(p-1)$
  Thus $g^{r+p-1+p-1} = g^r = y$ again by ~~Fermat~~.

I'm a bit confused

Hence we may reduce it to the case $3 \mid r$.
Then there exists a $d \in \mathbb{Z}$ s.t. $3 \cdot d = r$. By fixing $x = g^d \in \mathbb{F}_p^\times$, we get that

$$x \longmapsto x^3 = g^{3d} = g^r = y.$$

Hence, this map is surjective and therefore bijective so we can conclude. ∎

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